

Effective connectivity of emotion and cognition under psilocybin

Devon Stoliker^{1,2*}, Leonardo Novelli^{1,2}, Franz X. Vollenweider³, Gary F. Egan^{1,2}, Katrin H. Preller³⁺, Adeel Razi^{1,2,4,5+}

¹Turner Institute for Brain and Mental Health, Monash University, Clayton, VIC

² Monash Biomedical Imaging, Monash University, Clayton, VIC

³ Department of Psychiatry, Psychotherapy & Psychosomatics, Psychiatric University Hospital Zurich, Zurich, Switzerland

⁴ Wellcome Centre for Human Neuroimaging, UCL, London, United Kingdom

⁵ CIFAR Azrieli Global Scholars Program, CIFAR, Toronto, Canada

* Corresponding author:

Devon Stoliker, Monash Biomedical Imaging, 762-772 Blackburn Rd, Clayton VIC 3168. Australia.

Email: devon.stoliker@monash.edu

+ Joint Senior Authors

Word count = 4692

Abstract

Classic psychedelics alter sense of self and patterns of self-related thought. These changes are hypothesised to underlie their therapeutic efficacy across internalising pathologies such as addiction, anxiety, and depression. Using resting-state functional MRI images from a randomised, double blinded, placebo-controlled clinical trial of 24 healthy adults under 0.215mg/kg psilocybin, we investigated the effective connectivity changes between the amygdala and the default mode network (DMN), salience network (SN) and central executive network (CEN). This connectivity underpins the appraisal and regulation of emotion and is associated with clinical symptomatology. We observed a general pattern of decreased top-down effective connectivity from the resting state networks of interest to the amygdala and directed connectivity changes associated with altered emotion and meaning under psilocybin. Our

findings identify cognitive-emotional connectivity associated with the subjective effects of psilocybin and the attenuation of the amygdala as a potential biomarker of psilocybin's therapeutic efficacy.

Keywords – fMRI, psychedelics, resting state connectivity, dynamic causal modelling, psilocybin, emotion, cognition

Introduction

Psilocybin, the main active compound of so-called magic mushrooms, is a serotonergic psychedelic that has been used throughout history in religious and medical rituals to alter consciousness (Carod-Artal, 2015; Nichols, 2016). It belongs to a larger group of substances such as ayahuasca (DMT) and mescaline that share molecular binding kinetics to serotonergic 5-HT receptors distributed throughout the brain (Nichols, 2012). Agonism of the 5-HT_{2A} receptor is primarily responsible for psychedelic subjective effects that can elicit powerful changes to subjective experience (Vollenweider et al 1998, Kometer et al. 2013, Preller et al. 2017, Kraehenmann et al. 2017). These include profound alterations in self-consciousness and perception including visual illusions, hallucinations, and synaesthesia, emotion and cognitive capacities, and transcendence of space and time (Studerus, Gamma, & Vollenweider, 2010; Swanson, 2018; Wittmann et al., 2007; Stoliker, Egan, Friston, & Razi, 2021). Serotonergic psychedelic-induced alterations can also involve an experience of *ego dissolution* defined as a loss of self-boundaries and sense of unity between the self and the environment (Stoliker, et al., 2021; Lebedev et al., 2015). Although, complete ego dissolution is rare at the doses used in clinical trials, several studies have associated measurable degrees of ego dissolution (indexed as total mystical score or oceanic self-boundlessness) to symptom reduction of depression (Griffiths et al., 2016; Roseman, Nutt, & Carhart-Harris, 2018; Ross et al., 2016) and addiction (Bogenschutz et al., 2015; Garcia-Romeu, Griffiths, & Johnson, 2014). However, the cognitive and emotional neural mechanisms that underlie psychedelic subjective and therapeutic effects are not well understood. Therefore, our goal was to investigate

connectivity changes that may underlie the therapeutic efficacy of psychedelics and their association to behaviour.

Previous research has demonstrated change to the amygdala and associative connectivity under psychedelics. The amygdala's role in emotional arousal and the function of associative connectivity in cognition identify the amygdala connectivity as a target to investigate emotional and cognitive change under psilocybin. The amygdala is a bilateral subcortical region composed of many subnuclei that integrate cortical and thalamic sensory inputs and their emotional valence (Babaev, Piletti Chatain, & Krueger-Burg, 2018). For example, presentations of positive or negative emotional stimuli—such as physical threats—elicit activation of the amygdala (Alexander et al., 2021; Garavan, Pendergrass, Ross, Stein, & Risinger, 2001; Hobin, Goosens, & Maren, 2003; Weymar & Schwabe, 2016). The amygdala is also well known for its involvement in fear conditioning (Sun, Gooch, & Sah, 2020) and activates in experiences of emotional empathy (Geng et al., 2018; Hillis, 2014; Marsh et al., 2013). Hyperactivation of the amygdala is a marker of stress and anxiety and is implicated across internalising disorders for which psychedelics demonstrate efficacy (Nutt & Carhart-Harris, 2020; Koob & Volkow, 2010; Luo, Xue, Shen, & Lu, 2013; Menon, 2011). Interestingly, the amygdala also responds to beliefs (Kaplan, Gimbel, & Harris, 2016) suggesting that the amygdala conditions an emotional response to self-related beliefs that may underline internalising disorders. Damage to the amygdala affects decision-making (Gupta, Kosciak, Bechara, & Tranel, 2011), memory (Adolphs, Tranel, & Denburg, 2000), and behaviour regulation. Moreover, its damage can also diminish an individual's sense of shame (Piretti et al., 2020) further suggesting the amygdala mediates the relationship between emotion and cognition.

The amygdala is integrated with cognition through top-down projections from midline cortical regions (Etkin et al., 2011; Kraehenmann et al., 2016; Robinson et al., 2013) that underlie the perception and intensity of positive and negative emotions (Bonnet et al., 2015). This circuitry also underlies psychopathological conditions such as anxiety disorders (Menon, 2011). Modulation of top-down control

by psilocybin suggests influence on limbic region emotional processing (Bernasconi et al., 2014). For example, healthy adults administered psilocybin show reduced effective amygdala connectivity to the primary visual cortex, coinciding with reduced threat processing (Kraehenmann et al., 2016). The semisynthetic psychedelic drug lysergic diethylamide acid (LSD) also has been shown to dampen the amygdala response to the presentation of fearful faces (Mueller et al., 2017). Moreover, decreased amygdala activity has been correlated with positive mood under psilocybin (Kraehenmann et al., 2015). Together, these results suggest a decrease in amygdala connectivity in association with therapeutic response in healthy adults under the acute influence of psychedelics.

In contrast, clinical populations measured the day after administration of psilocybin demonstrated an opposite pattern of amygdala activity. The amygdala response to emotional faces has been reported to increase the day after psilocybin administration (Roseman, Demetriou, Wall, Nutt, & Carhart-Harris, 2018) and be accompanied by decreased functional connectivity with the prefrontal cortex (Mertens et al., 2020; Mueller et al., 2017). It has been proposed that increased amygdala activity in the sub-acute period may underlie a sense of emotional reconnection (Watts, Day, Krzanowski, Nutt, & Carhart-Harris, 2017). Changes to the amygdala and cortical regions are sustained well beyond the half-life of psilocybin and may last up to one month post administration (Barrett, Doss, Sepeda, Pekar, & Griffiths, 2020), demonstrating the lasting neural changes in association with positive responses to emotionally conflicting stimuli. These results indicate the acute effects of psychedelics in clinical studies may disrupt conditioned response patterns and produce enduring connectivity changes that support therapeutic modes of behaviour.

The importance of emotion and cognition to behaviour motivate investigation of cortical regions with established connections and associations to the amygdala. These include frontal and anterior regions that form nodes of distinct large-scale resting-state networks (RSNs). RSNs are determined by temporally correlated activation across spatially distributed brain regions and serve essential functions in cognition

(Fox et al., 2005; Raichle, 2015). The RSN regions of particular relevance to this study are abundant in 5-HT_{2A} receptors (R. L. Carhart-Harris & Friston, 2019). The functional integration of RSN nodes with the amygdala illustrate that emotion is a flexible and dynamic workspace within the brain (Lindquist, Satpute, Wager, Weber, & Barrett, 2016).

The default mode network (DMN) integrates internal and external information to perform self-referential mental activity that is closely identified with the sense of self. DMN hub regions comprise the medial prefrontal cortex (mPFC) and posterior cingulate cortex (PCC). The mPFC functions in temporally extended goal-directed thinking and regulation of behaviour, while the PCC orients information towards the self (Andrews-Hanna, Smallwood, & Spreng, 2014; J. A. Brewer, Garrison, & Whitfield-Gabrieli, 2013). Information integration in the DMN sustains self-evaluative processes involved in forming self-beliefs (Amey, Leitner, Liu, & Forbes, 2022; Tompson, Chua, & Kitayama, 2016).

DMN connectivity is related to wellbeing. For example, limbic-cortical dysregulation have been reported to impact mood regulation (Zeng et al., 2012) and increased FC between the hippocampus and prefrontal cortex (e.g., mPFC) was observed in depression (Kaiser, Andrews-Hanna, Wager, & Pizzagalli, 2015). Increased DMN connectivity with the amygdala is also observed in depressed subjects (Sheline Yvette et al., 2009) in association with symptoms such as rumination (Cooney, Joormann, Eugène, Dennis, & Gotlib, 2010). In contrast, decreased PCC-mPFC coupling is associated with the reduced mind wandering of advanced meditation practitioners (Judson A. Brewer et al., 2011) with similar decoupling found acutely (Carhart -Harris et al 2012) and the day after psilocybin (R. L.; Smigielski, Scheidegger, Komater, & Vollenweider, 2019).

DMN frontal regions, specifically the mPFC, share well-established connectivity to the amygdala that plays a role in the valence and regulation of emotion (Willinger et al., 2019; Underwood, Tolmeijer, Wibroe, Peters, & Mason, 2021; Cooney et al., 2010). The PCC node of the DMN is less often associated

with the amygdala. However, it plays a significant role in the experience of self. Under psychedelics, reduced PCC connectivity is associated with diminished feelings of self that align with descriptions of ego dissolution (R. L. Carhart-Harris et al., 2014) and reduced self-representation in social interaction (Preller et al., 2018). We hypothesise reduced interactions between the amygdala and DMN and reduced within-DMN connectivity may underwrite altered sense of self and self-related emotion under psychedelic subjective effects and serve as a biomarker of connectivity. that may be therapeutic.

Frontal regions also compose the dorsolateral prefrontal cortex (dlPFC) nodes of the central executive network (CEN). The CEN is a RSN involved in thinking, planning, and controlling attention (Andrews-Hanna et al., 2014; Menon, 2011). The dlPFC serves a primary role in regulating emotion (Underwood et al., 2021) and handles content in working memory and evaluates possible behavioural responses. The CEN also includes the lateral posterior parietal cortex (lPPC), which integrates sensory and interoceptive information to facilitate sustained attention (Borders, 2020). CEN functions are associated with mental health in various cognitive disorders. Hyper and hypo-frontal connectivity are often markers of disordered cognition and anxious behaviour. For example, increased CEN-AMG connectivity was observed in association with the catastrophising of chronic pain (Jiang et al., 2016), and dlPFC connectivity was associated with worry in healthy adults and individuals with generalised anxiety (Paulesu et al., 2010; Servaas et al., 2014; Jamieson, Harrison, Razi, & Davey, 2022). Moreover, reduced CEN activity has been associated with chronic schizophrenia and depression (Leptourgos et al., 2020; Menon, 2011). However, links between CEN activity and behaviour are suggested to be task and context-dependent (Borders, 2020). For example, reduced dlPFC was associated with reduced rumination observed in healthy adults (Cooney et al., 2010).

Under psychedelics, a number of brain mapping studies suggest CEN connectivity deviates from its usual functional pathways and proliferates to regions outside of this network. For example, LSD increases

global connectivity with the CEN (Luppi et al., 2021; Tagliazucchi et al., 2016), and psilocybin renders the phase-locking of blood oxygen level-dependent (BOLD) neurovascular signals flexible (Lord et al., 2019). These changes have been associated with the shift in perceptual boundaries and the reduced integrity of the brain's "rich-club" organisation (Tagliazucchi et al., 2016). However, RSN connectivity changes with the amygdala, such as the CEN, under psychedelics have not been well explored and may underlie change to cognitive-emotional regulation.

The salience network (SN) is a cognitive RSN involved in detecting stimuli associated with biological or goal-directed personal significance. Both the DMN and CEN are under the influence of the SN (Menon & Uddin, 2010; Sridharan, Levitin, & Menon, 2008). The SN recruits CEN activity to provide task-relevant information in response to detection of salient features in the environment and disengages task-irrelevant activity, for example, daydreaming. The SN's cardinal regions are the dorsal anterior cingulate cortex (dACC) and anterior insula (AI). The dACC is involved in response selection and conflict monitoring (Ide, Shenoy, Yu, & Li, 2013; Menon, 2011) and activated in response to social pain (Rotge et al., 2015). The AI detects behaviourally relevant stimuli (Menon, 2015) and receives multimodal sensory input (Averbeck & Seo, 2008; Vogt & Pandya, 1987). Expanded network compositions of the SN also include the amygdala, which may be important for the initial emotional appraisal of salient stimuli (Menon, 2011). Overactive salience attributed to stimuli is a feature of various forms of psychopathology. Hyperactivity of the AI or amygdala has been reported to underwrite enhanced salience detection (Menon, 2011). Moreover, increased salience and attention to the self during social interactions may underlie social anxiety, while overt salience and attention to negative stimuli or mentation has been suggested to reinforce depressed thinking. The salience of channels of attention (i.e., the top-down selection of bottom-up information that receives priority in attention) also plays a role in psychosis, suggesting the SN may act as pre-reflective mechanisms of attention that underlie perception and beliefs (Kapur, 2003).

Serotonergic receptors play a role in the selection of channels of attention (Stoliker, et al., 2021), and observations under psilocybin have shown reduced SN integrity (Lebedev et al., 2015). Changes to effective connectivity from the SN to the DMN are also observed under LSD (Stoliker, Novelli, et al., 2021). The dACC has been associated with changes in self-processing and a reduced sense of social exclusion under psilocybin administration that did not depend on changes in attention and awareness (Preller et al., 2016; Preller et al., 2015). However, amygdala links with the dACC and AI under psychedelics remain relatively unexplored and worthy of investigation for their contribution to the subjective effects and therapeutic benefits of psilocybin.

To address the gap in understanding emotion and cognition changes under psychedelics, we apply spectral dynamic causal modelling (spDCM) (Karl J. Friston, Kahan, Biswal, & Razi, 2014; Razi, Kahan, Rees, & Friston, 2015; Razi et al., 2017). DCM can infer the direction of connectivity between hierarchically organised brain regions composing RSNs (Zhou et al., 2018) and DCM has been used to measure the effective connectivity under LSD from the thalamus to the cortex (Preller et al., 2019) and among various RSNs (DMN, SN, DAN) (Stoliker, Novelli, et al., 2021). The ability of DCM to investigate bottom-up and top-down processes between regions of the brain suggests it as a valuable technique to investigate psilocybin-induced effective connectivity changes between high-level RSNs and the subcortical amygdala. Moreover, DCM allows the inference of association between effective connectivity changes and retrospective behavioural reports using the Bayesian framework of parametric empirical Bayes (PEB) (K. J. Friston et al., 2016). We measured *changed meaning of precepts* and *blissful state* collected 360 min following the administration of psilocybin using the five dimensions of altered states of consciousness scale (5D-ASC) (Studerus et al., 2010).

Previous evidence suggests that amygdala and RSN connectivity is decreased under psychedelics. The associative RSNs we examined express high density of the 5-HT_{2A}R that are thought to alter the top-down modulation of subcortical connectivity. Previous reports also suggest psilocybin can induce

psychological flexibility (Davis, Barrett, & Griffiths, 2020) and feelings of subjective wellbeing (Hirschfeld & Schmidt, 2020). We hypothesise reduced top-down effective connectivity from the RSNs to the amygdala under psilocybin. We anticipate effective connectivity changes in our model will be associated with altered meaning of precepts and blissful state. Evidence of cortex-amygdala effective connectivity changes associated to these behavioural changes is hypothesised to be a marker of connectivity that underlies psilocybin's capacity to facilitate therapeutic outcomes.

Results:

Network effective connectivity change with the amygdala under psilocybin

Participants underwent resting-state fMRI scans 70 minutes post oral administration of psilocybin. Effective connectivity between the amygdala and the DMN and CEN was analysed. The reported results for effective connectivity used a threshold of posterior probability > 0.99 for inference with very strong evidence. Those connections not reported in the subsequent figures did not exceed this threshold. For details of the intervals see Supplementary Table 1.

i) Change of DMN effective connectivity to the amygdala under psilocybin

Group level effective connectivity strength between regions of the DMN and amygdala showed decreased self-inhibition of the PCC and decreased effective connectivity from the PCC with the left amygdala compared to placebo. See Fig 1 and Supplementary for explanation of self-connections.

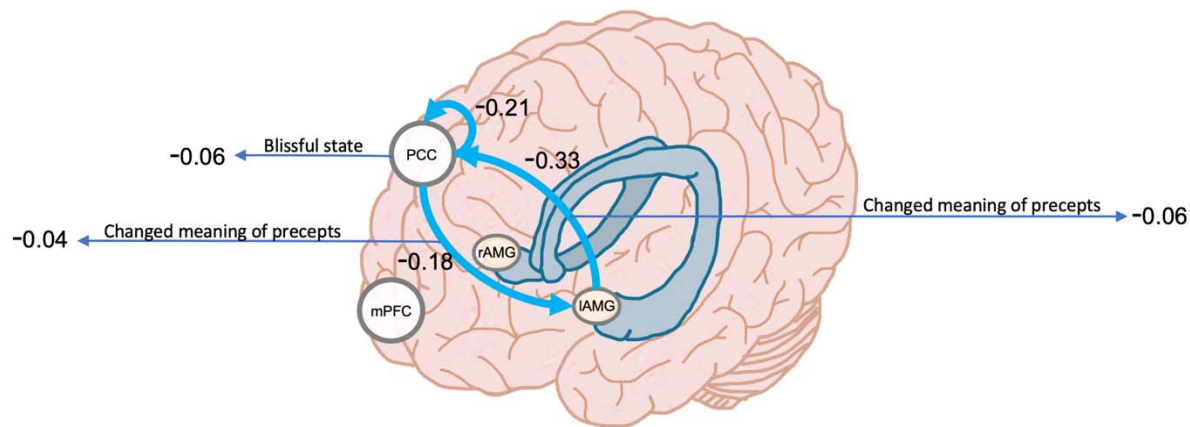


Fig 1. Default mode network effective connectivity change under psilocybin 70 minutes post-administration. Connections show changes in effective connectivity compared to placebo. PCC decreased self-inhibition (i.e., increased sensitivity to inputs) and connectivity to and from the left amygdala decreased compared to placebo. Values display effect sizes (posterior expectations) of connections in Hz (except the inhibitory self-connections which are log-scaled). All results are for posterior probability > 0.99.

ii) Change of CEN effective connectivity to the amygdala under psilocybin

Several changes in effective connectivity were found, whose interplay is captured schematically in Fig. 2. Specifically, group level effective connectivity strength between regions of the CEN and amygdala showed increased self-inhibition of the rdIPFC, ldIPFC and decreased self-inhibition of the ILPPC, compared to placebo. Effective connectivity from the ILPPC to the left amygdala decreased and increased to the rdIPFC and decreased to the right amygdala. ldIPFC effective connectivity to the right amygdala decreased and increased to the ILPPC, while rdIPFC increased to the ldIPFC and the right amygdala increased to the ILPPC, compared to placebo.

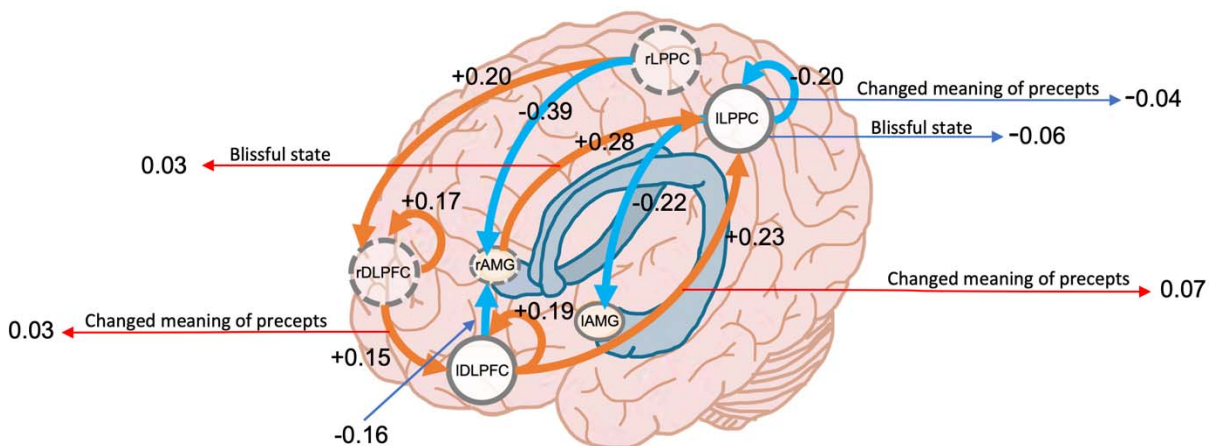


Fig 2. Central executive network effective connectivity change under psilocybin 70 minutes post-administration. Connections show changes in effective connectivity compared to placebo. rLPPC and IDLPFC connectivity to the right amygdala and ILPPC connectivity to the left amygdala become inhibitory compared to placebo. Bilateral self-connectivity of the dLPFC increased inhibition under psilocybin while the ILPPC self-connectivity decreased inhibition compared to placebo. These changes accompany excitatory change from the rLPPC to the rDLPFC, rDLPFC to the IDLPFC, the IDLPFC to the ILPPC and right amygdala to the ILPPC, compared to placebo. Values display effect sizes (posterior expectations) of connections in Hz (except the inhibitory self-connections which are log-scaled). All results are for posterior probability > 0.99.

iii) Change of SN effective connectivity to the amygdala under psilocybin

Group-level effective connectivity showed analysis showed increased self-inhibition of the rAI and bilateral amygdala, and decreased effective connectivity from the dACC and lAI to the right amygdala compared to placebo (Fig 3).

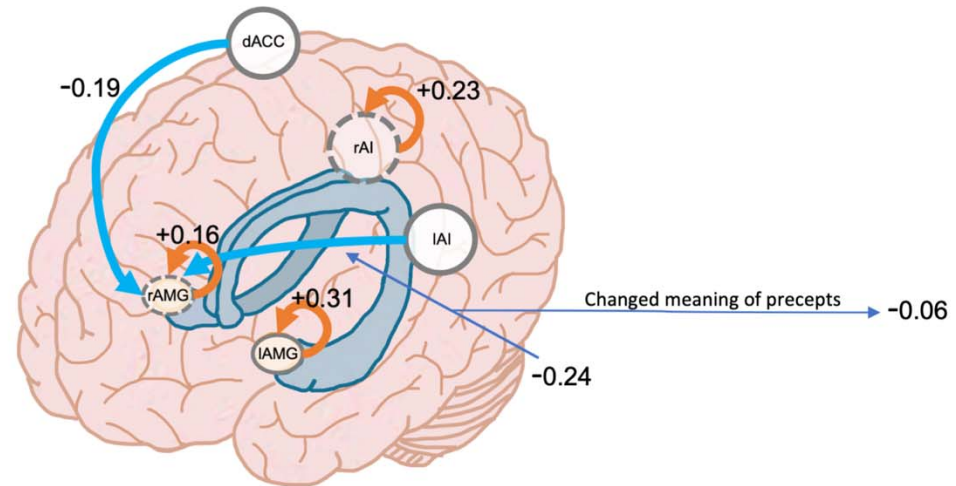


Fig 3. Salience network effective connectivity change under psilocybin 70 minutes post-administration. Connections show changes in effective connectivity compared to placebo. rAI and bilateral AMG self-inhibition increased, and connectivity from the dACC to the right AMG decreased compared to placebo. Values display effect sizes (posterior expectations) of connections in Hz (except the inhibitory self-connections which are log-scaled). All results are for posterior probability > 0.99.

Brain-behavior associations

i) Behavioural associations to DMN-amygdala effective connectivity change under psilocybin

Decreased self-inhibition of the PCC was associated with blissful state, while bidirectional PCC-left amygdala connectivity inhibition was associated with changed meaning of precepts. See Fig 1.

ii) Behavioural associations to CEN-amygdala effective connectivity change under psilocybin

Excitation of the right amygdala to the ILPPC was associated with blissful state. Changed meaning of precepts and blissful state were both associated with the decreased self-inhibition of the ILPPC.

Excitatory effective connectivity from the rdlPFC to the ldIPFC and ldPFC to lLPPC was associated with changed meaning of precepts. See Fig 2.

iii) *Behavioural associations to SN-amygdala effective connectivity change under psilocybin*

Inhibitory connectivity from the lAI to the right amygdala was associated with changed meaning of precepts. See Fig 3.

Discussion:

To investigate the influence of psilocybin on the interaction between emotion and cognition, we measured connectivity changes between the amygdala and selected RSNs. These networks serve functions in selfhood and cognition and are composed of regions linked to the regulation and experience of emotion. Our results demonstrate increased inhibitory effective connectivity from cortical regions to the amygdala under psilocybin. For example, we showed this change in connectivity from the PCC, ldPFC, bilateral lPPC, ACC, and lAI. These results suggest cognitive resources do not recruit the amygdala under psilocybin, and this may relate to subjective and therapeutic effects of psilocybin. While we did not find effective connectivity change involving the mPFC—which has been frequently cited in association with emotion, such as emotional valence (Willinger et al., 2019)—examination of the PCC and amygdala revealed multiple connectivity changes and behavioural associations under psilocybin. PCC and amygdala effective connectivity change was bidirectional and accompanied by reduced PCC self-inhibition, which indicates increased synaptic gain or its sensitivity to inputs. Increased PCC synaptic gain was associated with the subjective experience of blissful state, and PCC bilateral inhibition with the left amygdala was associated with changed meaning of precepts. Our findings suggest inhibition between amygdala-mediated emotion and self-referential functions as an important factor in these subjective changes that may contribute to the joyous feelings of self-dissolution of ego dissolution and reduced self-other

processing under psychedelics (Preller et al., 2018; R. Carhart-Harris, Kaelen, & Nutt, 2014; Stoliker, Novelli, et al., 2021).

IDLPC and bilateral ILPPC inhibition to the amygdala is situated within the CEN. Connectivity in this network is believed to underlie the evaluative aspect of thought, with the DLPFC playing a central role in emotional regulation. However, IDLPFC inhibition to the amygdala was not associated with subjective effects. Instead, the pattern of activity in this network is open to interpretation. One characterisation of the observed activity may be its resemblance to hyper-frontality, which is associated with worry and anxiety (Park, Wood, Bondi, Del Arco, & Moghaddam, 2016), and acute psychosis (Vollenweider et al., 1997). Participants in this study were experiencing the effects of psychedelics in the setting of an fMRI scan, which may arouse substantial anxiety. Moreover, strong parallels between cortical activity patterns associated with psychedelic subjective effects have been shown to resemble anxiety (Stoliker, et al., 2021). However, subjects of this study reported nominal anxiety, see Supplementary Material. An alternative interpretation is that the observed connectivity pattern between regions of the CEN constitutes an increase in evaluative thought processes. This interpretation aligns with the function of the CEN and evidence that increased intrinsic CEN connectivity is related to resilience (Miller Gregory et al., 2018). This may suggest the within-network excitatory effective connectivity change as a biomarker for potential therapeutic revision of beliefs.

Although we expect re-evaluations of thought under psychedelics to be influenced by the context, measured subjective effects associated with CEN connectivity support this view. For example, increased connectivity from the IDLPFC to ILPPC and connectivity between the bilateral DLPFC was associated with changed meaning of precepts. Moreover, decreased ILPPC self-inhibition and its directed interconnectivity with the IDLPFC and rAMG constituted a primary source of behavioural change related to ego dissolution. The bilateral ILPPC inhibition to the bilateral amygdala suggests its decoupling from the amygdala limiting the sensory and interoceptive information to the amygdala, while the left amygdala

synaptic gain increased. These results align with previous reports of increased flexibility in thinking, functional desegregation and expansion of brain states made available under psychedelics which may be important to the therapeutic experience under psilocybin and lasting cognitive-emotional changes (Atasoy et al., 2017; Barnett, Muthukumaraswamy, Carhart-Harris, & Seth, 2019; R. L. Carhart-Harris & Nutt, 2017; Davis et al., 2020; Matias, Lottem, Dugué, & Mainen, 2017; Stoliker, Egan, & Razi, 2022; Tagliazucchi, Carhart-Harris, Leech, Nutt, & Chialvo, 2014; Watts & Luoma, 2020). Moreover, previous brain network research on emotion indicates that flexible and nimble brain networks are instrumental to positive emotion and affect (Alexander et al., 2021).

The ACC and IAI serve essential functions in salience detection, and these regions' integration with the amygdala may underwrite pathologically enhanced salience detection (Menon, 2011). These regions inhibition to the right amygdala suggests their role in salience detection is diminished under psilocybin, which may be therapeutic in internalising disorders characterised. However, inhibited salience detection also aligns with positive symptoms in schizophrenia (i.e., hallucinations and delusions) (Menon, 2011). At higher doses, psychedelics can induce a psychotic-like state in some individuals that may be related to the effect of serotonergic psychedelics on salience detection. However, psychotic-like behavioural outcomes rarely occur in a controlled clinical setting.

Previous research shows the increased functional connectivity between the AI and amygdala is associated with relapse of addiction in preclinical studies (Venniro et al., 2017) and behavioural habituation in healthy adults (Denny et al., 2014). The AI inhibition to the amygdala was associated with changed meaning of precepts suggesting it may be associated with therapeutic behavioural change. Previous task-based research shows reduced dACC activity in association with blunted response to social exclusion under psilocybin. Although our findings did not associate subjective effects with this connection, inhibition from the dACC to the right amygdala aligns with the inhibition of social pain (Preller et al., 2016; Preller et al., 2015) and previous findings of this circuits top-down inhibition in the resolution of

emotional conflict (Etkin, Egner, Peraza, Kandel, & Hirsch, 2006). DCM also estimated increased self-inhibition in the bilateral amygdala and rAI, suggesting reduced synaptic gain of the regions. However, these changes were not associated with the measured psychedelic subjective effects.

Taken together, our results suggest that psilocybin attenuates the recruitment of the amygdala. This connectivity contributes to the altered meaning of precepts and subjective emotion. Diminished amygdala activation may explain reduced aversion to emotional stimuli and may alter the patterns of connectivity that condition emotional, cognitive and self-reflective responses of the brain. The RSNs reduced connectivity to the amygdala and altered within-RSN effective connectivity may be important conditions for ego dissolution and emotional-cognitive changes that facilitate the therapeutic revision of beliefs and therapeutic outcomes following psilocybin assisted therapy.

Our endeavour to investigate interactions between emotion and cognition in a healthy population under psilocybin has several limitations. For example, our analysis would benefit from MRI scans taken before and after the administration of psilocybin. Exploring links between emotional and associative connectivity also involves deciding network configuration and regions for analysis. Regions such as the ventral striatum, precuneus, and parahippocampus and further subregions of the PFC and ACC may also be valuable for examining the links between emotion and cognition (Underwood et al., 2021). Furthermore, each region selected as part of the network may be determined differently depending on the method of identifying the region. Further standardisation of this process is complex for anatomical reasons and remains under debate. The processing pipeline, participant sample and dose can also strongly impact the reliability of results. Participants were given less than currently used in clinical studies into depression and substance dependence disorders. Therefore, connectivity dynamics may be dissimilar at a higher dose. Data sets of higher doses may provide improved inference and more nuanced characterisation of neural dynamics. We also overlooked the influence of numerous neurotransmitter

systems involved in emotion that may interact with altered serotonin transmission. These interactions may provide a novel line of inquiry to understand psychedelic therapeutic effects.

Other limitations in the methods and analysis include the use of global signal regression and the limited temporal resolution. Temporal resolution may be significant to emotion. For example, the time course of emotion may present distinct contributions of brain regions involved in valence and arousal (Alexander et al., 2021). Modalities with temporal specificity capable of resolving insight and emotional change, such as EEG or MEG, which can also be analysed using DCM, are needed in future investigations. Furthermore, studies using task-based fMRI paradigms that can induce activations in specific emotion and cognition related brain regions.

The generalisability of our results to clinical populations is limited by the use of a small healthy adult sample and the congruence between our results and previous findings. Previous research indicates psychedelics may globally increase bottom-up connectivity (R. Carhart-Harris, Kaelen, & Nutt, 2014; R. L. Carhart-Harris & Friston, 2019; Swanson, 2018). For example, evidence of thalamic connectivity to the cortex supports this view (Müller, Dolder, Schmidt, Liechti, & Borgwardt, 2018; Preller et al., 2019). However, decoupling between limbic regions and the cortex has also been identified (Robin L. Carhart-Harris et al., 2016; Lebedev et al., 2015). Our evidence only found increased bottom-up connectivity from the right amygdala to the lIPPC and aligns with the latter studies. As psychedelic research advances, we suggest examination of multiple limbic regions and their subnuclei connectivity to the cortex. This may reveal more precise and nuanced dynamics between the subcortex and cortex underlie previously theorised global patterns. For example, recent investigations have demonstrated spatial specificity of entropic activity and thalamic connectivity associated with psychedelic subjective effects (Gaddis et al., 2022; Herzog et al., 2020). Along these lines, the amygdala is composed of many subnuclei. Its small size, anatomical variability and position deep within the brain presents challenges for noise-to-signal ratio and spatial location in group fMRI analysis, which future imaging and research methods may overcome.

Finally, increased directed excitation among the CEN contrasts with previous analyses, which found a general reduction in associative functional connectivity (Katrin H. Preller et al., 2020). While the global reduction of associative connectivity may help explain experiences of ego dissolution, psychedelic experiences also contain phenomenological richness (Millière, Carhart-Harris, Roseman, Trautwein, & Berkovich-Ohana, 2018). Future research of CEN regions may wish to address the reliability of this directed connectivity and examine links between the CEN, subjective experience, and therapeutic outcomes.

In sum, we demonstrated altered within-RSN effective connectivity and reduced top-down RSN-amygdala effective connectivity associated with emotional and cognitive behavioural changes under psilocybin. The functional significance of this connectivity in perceptual processes and psychopathology suggests its acute alteration by psilocybin may elicit changes to conscious perception that are important to subjective wellbeing. Psilocybin altered cognition underwritten by connectivity changes may contribute to lasting therapeutic outcomes and enable the emotional and cognitive flexibility that facilitate alternate selection of perceptual hypotheses which underwrite self-beliefs. In particular, we suggest RSN attenuation of the amygdala as a potential biomarker for future clinical research investigating the neural correlates of psychedelic therapeutic efficacy. Future work may also build upon our directed connectivity findings to understand how connectivity involved in emotion and cognition interact with the therapeutic context to effect patterns of thought and behaviour. These lines of enquiry can help discern the imperative of the psychedelic subjective experience to therapeutic outcomes and bridge gaps in our understanding of the relationship between brain connectivity and behaviour.

Methods

Participants

The data analysed in this paper were collected as part of a previous study (registered at ClinicalTrials.gov (NCT03736980)), which is reported in (Katrin H. Preller et al., 2020) and was approved by the Cantonal Ethics Committee of Zurich. Twenty four subjects (12 males and 11 females; mean age = 26.3 y; range = 20–40 y; 1 subject did not complete the study) were recruited through advertisements at universities in Zurich, Switzerland. All participants were deemed healthy after screening for medical history, physical examination, blood analysis, and electrocardiography. The Mini-International Neuropsychiatric Interview (MINI-SCID) (Sheehan et al., 1998), the DSM-IV fourth edition self-rating questionnaire for Axis-II personality disorders (SCID-II) (Fydrich, Renneberg, Schmitz, & Wittchen, 1997), and the Hopkins Symptom Checklist (SCL-90-R) (Franke, 2002) were used to exclude subjects with present or previous psychiatric disorders or a history of major psychiatric disorders in first-degree relatives. Participants were asked to abstain from prescription and illicit drug use two weeks prior to first testing and throughout the duration of the study, and abstain from alcohol use 24 hours prior to testing days. Urine tests and self-report questionnaires were used to verify the absence of drug and alcohol use. Urine tests were also used to exclude pregnancy. Further exclusion criteria included left-handedness, poor knowledge of the German language, cardiovascular disease, history of head injury or neurological disorder, history of alcohol or illicit drug dependence, MRI exclusion criteria, including claustrophobia, and previous significant adverse reactions to a hallucinogenic drug. All participants provided written informed consent statements in accordance with the declaration of Helsinki before participation in the study. Subjects received written and oral descriptions of the study procedures, as well as details regarding the effects and possible risks of drug treatment.

Design

A double blind, randomised, placebo-controlled, cross-over study was performed. Testing days occurred two weeks apart and participants were orally administered either psilocybin (0.2mg/kg orally) or placebo (179 mg mannitol and colloidal silicon dioxide (Aerosil) 1 mg. Resting state scans (10 minutes each) were taken 20, 40 and 70 minutes following administration. However, only scans at 70 minutes during the

peak effects were used in our analysis. Participants were asked to not engage in repetitive thoughts such as counting and instructed to close their eyes during the resting state scan. Subjective experience was assessed using the retrospective 5-Dimensions Altered States of Consciousness (5D-ASC) questionnaire. Participants completed the 5D-ASC 360 min following psilocybin administration, see Supplementary.

MRI Data Acquisition and Preprocessing

MRI data were acquired on a Philips Achieva 3T whole-body scanner. A 32-channel receive head coil and MultiTransmit parallel radio frequency transmission was used. Images were acquired using a whole-brain gradient-echo planar imaging (EPI) sequence (repetition time, 2,430 ms; echo time, 27 ms; slice thickness, 3 mm; 45 axial slices; no slice gap; field of view, $240 \times 240 \text{ mm}^2$; in-plane resolution, $3 \times 3 \text{ mm}^2$; sensitivity-encoding reduction factor, 2.0). 265 volumes were acquired per resting state scan resulting in a scan duration of 10 mins. Additionally, two high-resolution anatomical images (voxel size, $0.7 \times 0.7 \times 0.7 \text{ mm}^3$) were acquired using T1-weighted and T2-weighted sequences using 3D magnetization prepared rapid-acquisition gradient echo sequence (MP-RAGE) and a turbo spin-echo sequence, respectively. See Supplementary for more details. The acquired images were analysed using SPM12 (<https://www.fil.ion.ucl.ac.uk>).

The pre-processing steps of the images consisted of slice-timing correction, realignment, spatial normalization to the standard EPI template of the Montreal Neurological Institute (MNI), and spatial smoothing using a Gaussian kernel of 6-mm full-width at half maximum. Head motion was investigated for any excessive movement. Three subjects were excluded due to head motion and one subject did not complete the scan at 70 minutes.

Independent RSN-AMG DCM models

To reduce model complexity and focus the interpretation of the results on the connections of interest our analysis investigated independent models of amygdala effective connectivity with the DMN, CEN and SN.

487

488 *Extraction of region coordinates across subjects*

489 Neurosynth, a large-scale automated synthesis of functional neuroimaging data, was used to identify

490 coordinates composing regions of interest (Yarkoni, Poldrack, Nichols, Van Essen, & Wager, 2011).

491 More details can be found in Supplementary.

492 Networks were composed of cardinal regions constituting core parts of the DMN (Andrews-Hanna,

493 Reidler, Sepulcre, Poulin, & Buckner, 2010; Dixon et al., 2017), SN and CEN. The DMN and SN

494 followed the selection of regions in a related investigation by Zhou and colleagues (Zhou et al., 2018) and

495 all networks were included for their role in consciousness and link to emotion (Alexander et al., 2021;

496 Borders, 2020; Underwood, Tolmeijer, Wibroe, Peters, & Mason, 2021). Associations between peak

497 coordinates and cardinal nodes of network regions of interest (ROI) were determined by expert visual

498 inspection. The MNI coordinates of the selected ROIs are listed in Table 1.

499 A generalized linear model (GLM) was used to regress 6 head motion parameters (3 translation and 3

500 rotational), white matter and cerebrospinal fluid signals from preprocessed data. One subject was

501 excluded as no activation was found in one or more regions of interest. We also used global signal

502 regression in our pre-processing pipeline in line with our previous work (Stoliker, Novelli, et al., 2021).

503 The time series for each ROI was computed as the first principal component of the voxel activity within a

504 6 mm radius sphere centred on the ROI coordinates (as listed in Table 1).

<u>Region</u>	<u>MNI coordinates</u>			<u>Network/System</u>
	x	y	z	
PCC	0	-50	28	DMN
mPFC	0	54	20	DMN
ACC	2	28	22	SN
IAI	-34	19	0	SN
rAI	38	18	0	SN
ldlPFC	-46	30	32	CEN
rdlPFC	42	37	31	CEN
lIPPC	-22	-62	62	CEN

<u>Region</u>	<u>MNI coordinates</u>			<u>Network/System</u>
	x	y	z	
rIPPC	19	-64	55	CEN
lAMYGDALA	-21	-5	-17	LIMBIC
rAMYGDALA	23	-3	-19	LIMBIC

505
506 *Table 1 Coordinates of regions of interest. The DMN comprised of the posterior cingulate cortex (PCC)*
507 *and medial prefrontal cortex (mPFC); the SN comprised of the dorsal anterior cingulate cortex (dACC),*
508 *left and right anterior insula (lAI/rAI); and the CEN comprised the left and right dorsal lateral prefrontal*
509 *cortex (rdlPFC/rdlPFC) and the left and right lateral posterior parietal lobule (lIPPC/rIPPC).*

510
511 *Specification and Inversion of DCM*
512 Three independent fully-connected DCM models were specified for each network and the amygdala using
513 the 11 ROIs defined in Table 4, without any exogenous inputs. The DCM for each subject was then
514 inverted using spectral DCM (Karl J. Friston, Kahan, Biswal, & Razi, 2014; Razi, Kahan, Rees, &
515 Friston, 2015) to infer the effective connectivity that best explains the observed cross-spectral density for
516 each subject. This procedure was repeated for each testing condition. The DCM fit to the data using cross
517 spectral density averaged 90.9% for placebo conditions and 89.8% for psilocybin conditions.

518
519 *Second Level Analysis Using Parametric Empirical Bayes*
520 The effective connectivity inferred by spectral DCM for each subject was taken to the second (group)
521 level to test hypotheses concerning between-group effects. A general linear model (GLM) was employed
522 to decompose individual differences in effective connectivity into hypothesised group-average connection
523 strengths together with unexplained noise. Hypotheses on the group-level parameters are tested within the
524 parametric empirical Bayes (PEB) framework (K. J. Friston et al., 2016), where both the expected values
525 and the covariance of the parameters were taken into account. That is, precise parameter estimates

influence the group-level result more strongly than uncertain estimates, which are down-weighted. Bayesian model reduction (BMR) is used as an efficient form of Bayesian model selection (K. J. Friston et al., 2016). Please see supplementary material for further technical details about PEB.

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